Lab Assignment: Shortest Remaining Time First (SRTF) CPU Scheduling Algorithm

Objective:

To understand and implement the SRTF CPU scheduling algorithm and compute:
- Gantt Chart
- Completion Time (CT)
- Turnaround Time (TAT)
- Waiting Time (WT)
- Average Turnaround Time
- Average Waiting Time

Reference:

Step 1: Input
1. Read number of processes n
2. For each process i:
o Read Arrival Time AT[i]
o Read Burst Time BT[i]
o Set Remaining Time RT[i] = BT[i]

Step 2: Initialize
3. Set:
o time = 0
o completed = 0
o min_remaining_time = infinity

Step 3: Scheduling Process
4. While completed < n, repeat:
1. Find process j such that:
AT[j] ≤ time
RT[j] > 0
RT[j] is minimum among all available processes
2. If no such process is found:
Increment time
Continue to next iteration
3. Else:

Execute process j for 1 time unit
Decrement RT[j] by 1
Increment time by 1
4. If RT[j] == 0:
completed = completed + 1
Calculate:
Completion Time (CT[j]) = time
Turnaround Time (TAT[j]) = CT[j] − AT[j]
Waiting Time (WT[j]) = TAT[j] − BT[j]

Step 4: Output
5. Calculate:
o Average Waiting Time
o Average Turnaround Time
6. Display:
o Process ID
o Arrival Time
o Burst Time
o Waiting Time
o Turnaround Time

```java
import java.util.Scanner;

public class SRTF {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter number of processes: ");
        int n = sc.nextInt();

        int[] at = new int[n];
        int[] bt = new int[n];
        int[] rt = new int[n];
        int[] ct = new int[n];
        int[] tat = new int[n];
        int[] wt = new int[n];

        for (int i = 0; i < n; i++) {
            System.out.println("Process " + (i + 1) + ":");
            System.out.print(s: "Arrival Time: ");
            at[i] = sc.nextInt();

            System.out.print(s: "Burst Time: ");
            bt[i] = sc.nextInt();

            rt[i] = bt[i];
        }

        int completed = 0, time = 0;
        int minIndex = -1;
        int minRemaining;
        boolean found;
```

```java
while (completed < n) {

    minRemaining = Integer.MAX_VALUE;
    found = false;

    for (int i = 0; i < n; i++) {
        if (at[i] <= time && rt[i] > 0 && rt[i] < minRemaining) {
            minRemaining = rt[i];
            minIndex = i;
            found = true;
        }
    }

    if (!found) {
        time++;
        continue;
    }

    rt[minIndex]--;
    time++;

    if (rt[minIndex] == 0) {
        completed++;
        ct[minIndex] = time;
        tat[minIndex] = ct[minIndex] - at[minIndex];
        wt[minIndex] = tat[minIndex] - bt[minIndex];
    }
}

double totalWT = 0, totalTAT = 0;
```

```java
        System.out.println(x: "\nProcess\tAT\tBT\tCT\tTAT\tWT");

        for (int i = 0; i < n; i++) {
            totalWT += wt[i];
            totalTAT += tat[i];

            System.out.println("P" + (i + 1) + "\t" +
                    at[i] + "\t" +
                    bt[i] + "\t" +
                    ct[i] + "\t" +
                    tat[i] + "\t" +
                    wt[i]);
        }

        System.out.println("\nAverage Waiting Time = " + (totalWT / n));
        System.out.println("Average Turnaround Time = " + (totalTAT / n));

        sc.close();
    }
}
```

```
Enter number of processes: 5
Process 1:
Arrival Time: 0
Burst Time: 5
Process 2:
Arrival Time: 2
Burst Time: 3
Process 3:
Arrival Time: 4
Burst Time: 1
Process 4:
Arrival Time: 6
Burst Time: 2
Process 5:
Arrival Time: 8
Burst Time: 4

Process AT      BT      CT      TAT     WT
P1      0       5       5       5       0
P2      2       3       11      9       6
P3      4       1       6       2       1
P4      6       2       8       2       0
P5      8       4       15      7       3

Average Waiting Time = 2.0
Average Turnaround Time = 5.0
```

```
Enter number of processes: 5
Process 1:
Arrival Time: 0
Burst Time: 3
Process 2:
Arrival Time: 6
Burst Time: 4
Process 3:
Arrival Time: 7
Burst Time: 2
Process 4:
Arrival Time: 15
Burst Time: 5
Process 5:
Arrival Time: 16
Burst Time: 3

Process AT      BT      CT      TAT     WT
P1      0       3       3       3       0
P2      6       4       12      6       2
P3      7       2       9       2       0
P4      15      5       23      8       3
P5      16      3       19      3       0

Average Waiting Time = 1.0
Average Turnaround Time = 4.4
```

```
Enter number of processes: 5
Process 1:
Arrival Time: 0
Burst Time: 7
Process 2:
Arrival Time: 0
Burst Time: 4
Process 3:
Arrival Time: 0
Burst Time: 1
Process 4:
Arrival Time: 0
Burst Time: 3
Process 5:
Arrival Time: 0
Burst Time: 2

Process AT      BT      CT      TAT     WT
P1      0       7       17      17      10
P2      0       4       10      10      6
P3      0       1       1       1       0
P4      0       3       6       6       3
P5      0       2       3       3       1

Average Waiting Time = 4.0
Average Turnaround Time = 7.4
```